

Andrey Dodonov

Advanced Software Engineer

CONTACT

aadodonov@gmail.com
github.com/AndreyDodonov-EH
linkedin.com/in/andrey-dodonov

SKILLS

Systems & Embedded

C/C++, Rust,
Zephyr RTOS,
STM32, NRF,
SIL3 & Ex-safety,
NB-IoT, LTE-M, Wi-Fi,
BLE, Ethernet

Web

Lit, React, Vite,
JS/TS, Node.js,
REST, WS, HTTP,
TCP/IP

Blockchain

Polkadot-API, Polkadot SDK, JAM,
EVM, Solidity

Test & DevOps

Python, C#,
Docker, containerized development,
Jenkins, GitHub Actions

EDUCATION

BaselHack 2024 — Winner
hackathon · 2024

Polkadot Blockchain Academy (PBA)
Campus
cum laude
Hochschule Luzern (HSLU) · 2025

PBA-X academy
online course · 2025

M.Sc. Computer Science &
Engineering
cum laude
Southern Federal University
2008 – 2014

LANGUAGES

English	proficient
German	advanced
Russian	proficient

Advanced Software Engineer

Embedded · Blockchain · Full Stack · Industrial IoT

PROFILE

12+ years of experience designing systems with constrained resources, specializing in embedded, web development and blockchain technologies. Deep understanding of low-level protocols and efficient resource utilization. Programmed client and server web devices with under 1 Megabyte footprint. Filed patent on transferring web-pages over arbitrary non-IP protocols. Wrangling AI since GPT 3. Polkadot Blockchain Academy Campus graduate (cum laude).

EXPERIENCE

Advanced Software Engineer

2014 – Present

Endress+Hauser Group · Germany

- **Family of web-connected devices for liquid analysis**
 - Enabled rich Web UI for tiny devices (patent filed, architecture supports sockets of arbitrary protocols on non-IP direct links, allowing HTTP/WS interfaces for devices without TCP/IP stack)
 - Open-source contributions to the underlying Zephyr RTOS, specializing in networking
 - Added security features, CRA compliance, successfully surviving pen-test and ensuring zero security incidents in the field
 - Supported integration of industrial IP-based protocols
- **C Compiler for custom ASIC**
Fixed bugs and reduced output footprint of proprietary C compiler for custom ASIC
- **Organisational improvements**
 - Introduced containerized development with embedded devices, allowing quick on-boarding for new members, eliminating works-on-my-machine and mitigating supply chain attacks
 - Reduced vendor-lock on multiple fronts (compilers, version control, IDEs etc.)
 - Migrated company-wide front-end component library from React to Lit, reducing average front-end size 10 times from 2Megabytes to 200KBytes, enabling full-featured front-end for devices with limited resources
 - Pushed company-wide (1000+ users) migration to Git from legacy systems
 - Contributed to open-source Git wrappers, including popular GitExtensions
- **Augmented reality prototype for HoloLens for industries**
Integrating Unity, low-level protocols and HoloLens features, allowing to look into storage facilities and vessels as if they were transparent, based on Bluetooth, Wi-Fi and 3D mesh
- **Prototypes of embedded devices with various connectivity**
Prototyped devices with communication NB-IoT, LTE-M, Wi-Fi, BLE with memory footprints of under 1 MB, selected most efficient and secure approaches for systems
- **Cloud platform**
Integrated cloud services, web pages and sensors into coherent systems, performed migrations to cheaper hostings (from Heroku to GCP, Azure and on-prem), optimized data-heavy algorithms for cloud services (deep-dive from JS sorting to JS engines to underlying C++ libraries to cache locality)
- **Safety critical oil and gas sensor**
Hired as "fireman", quickly onboarded, fixed multiple complex bugs spread across 10 embedded chips in a single system, implemented safety-critical features and tests, allowing SIL3 certification

Freelance Consultant — Blockchain, Web & Mobile

2016 – Present

BusyBytes GmbH · Germany

- **Prototyped stablecoin smart contracts**
Developed and compared implementations across multiple blockchains, studied legal and regulatory aspects
- **Staking**
Hardened Linux environments for blockchain staking
- **Real estate management app (ImmoApp)**
Added real-time database synchronization, allowing onboarding of clients with legacy systems
- **Smart biking lamp with BLE (Lupine)**
Improved lamp to app protocol, making communication robust and user-friendly